

Rating Formula 1

Joint order statistics, a disaster slab,
and seventy years of mechanical attrition

Peter Cotton

July 9, 2026

Abstract

Formula 1 is a hostile environment for rating systems: grids of ten to fifty-five cars, full finish orders, and a retirement process that historically eliminated half the field. We rate every world-championship grand prix from 1950 to the present (1,158 races) under a prequential protocol — predict, score, then update — using a lattice-based Thurstonian system whose beliefs are unrestricted densities and whose update uses the exact likelihood of the whole finish order, computed in $O(N)$ per contestant. It beats TrueSkill, Elo, Glicko-2 and OpenSkill on log loss, accuracy and rank correlation, while the pairwise-decomposition systems degrade or diverge as grids widen: Glicko-2 scores below the uniform baseline. Because the lattice imposes no distributional family, the performance density is a modeling choice, and the right choice is not Gaussian: adding a *disaster slab* — a lump of far-slow mass matching the retirement process — improves every metric, and letting the slab mass track the trailing retirement rate sets the overall record while its fitted value follows the sport’s reliability revolution from 59% attrition in the 1980s to 13% today. We report what failed candidly: symmetric heavy tails hurt (they invent impossibly fast miracles), skew does nothing anywhere, and a near-deterministic pace core is refuted — pace noise among finishers is genuinely order one. What survives is a taxonomy: model catastrophe frequency, not time-domain shape. Against 135 races of archived betting odds the market wins by 0.57 of log loss and a fitted pool gives the ratings zero weight: Formula 1 win markets price the present — qualifying, upgrades, weather — and no compression of the past competes.

1 Why Formula 1

Most rating systems were built for head-to-head play and extended to multi-entrant events by decomposition: a finish order becomes a bag of pairwise results (Elo, Glicko-2, the Weng–Lin variants) or a chain of adjacent factors (TrueSkill). At tennis scale the decomposition is harmless. At Formula 1 scale it is the whole question. A grid of twenty cars implies 190 pairwise results whose errors are far from independent, and the sport’s history includes grids of fifty-five.

Formula 1 also embeds a distributional regime change inside a single dataset. The fraction of entrants failing to finish, by decade:

decade	1950s	60s	70s	80s	90s	2000s	10s	20s
DNF rate	.42	.49	.51	.59	.51	.30	.18	.13

No fixed distributional assumption is right for both halves of this table. A rating system whose noise model is a free parameter can say so; one locked to a Gaussian cannot.

The data (the community-maintained fldb compilation, CC-BY) is clean about edge cases that matter: drivers stopping past 90% distance are classified with positions; earlier retirements carry no position and enter our protocol tied at the back of the field; shared drives in the 1950s produce genuine dead heats, which the rater treats as ties rather than breaking arbitrarily.

2 The rater

Each driver carries a belief over latent ability: a whole density on a uniform lattice, free to be skewed or multimodal, diffusing between races in proportion to elapsed time. A race performance is ability plus noise drawn from a *base density* — any density; this freedom is the paper’s subject. Predictions price the joint order statistics of the field exactly (dead-heat aware, $O(N)$ per race) using the machinery of the fast ability transform. Updates use the exact likelihood of the observed finish order in each driver’s ability, with opponents at their predictive marginals: with finishers indexed fastest first, a forward chain accumulates prefix sums of (density $\times$ chain) over the faster finishers, a backward chain does the same over the slower ones, and the k -th finisher’s likelihood is the base density correlated with the product of the two chains. One coherent expression per driver, $O(N)$ overall, validated against brute-force simulation. An earlier version of the system factorized the order into peeled stages and multiplied overlapping factors; the chain replaced it after a paired test showed the factorization was measurably lossy — worse everywhere, at twice the cost.

3 The benchmark

All systems observe identical data prequentially: the first 20% of races are unscored warm-up; every later race is predicted before it is observed. Log loss, Brier and accuracy score the winner prediction; Kendall’s tau the predicted order; expected calibration error (ECE) the pooled win probabilities; and a randomized rank-PIT statistic measures whether each system’s *distribution* over finish positions matches reality. TrueSkill receives the empirical dead-heat rate; comparison rows use each package’s own prediction formula.

system	log loss	accuracy	tau	ECE	rank-PIT
Thurstone lattice	2.225	.316	.328	.015	.176
TrueSkill	2.346	.275	.300	.011	.199
Elo (multi-entrant)	2.465	.303	.319	.026	.220
OpenSkill Plackett–Luce	2.705	.259	.283	.023	.223
OpenSkill Thurstone–Mosteller	2.951	.007	.106	.010	.628
uniform baseline	3.162	.043	—	—	—
Glicko-2	3.727	.271	.255	.022	.166

Two failure modes deserve notice. Glicko-2, sound at head-to-head scale, scores *below the uniform baseline*: flooding a single rating period with nineteen correlated pairwise results drives its volatility machinery to numerical extremes (our implementation already includes stabilizing guards; without them it does not finish the century). OpenSkill’s Thurstone–Mosteller variant, which decomposes the order into all pairs, diverges similarly. The lesson is structural: pairwise decomposition is not a harmless approximation at grid scale.

4 The performance density is a free parameter

Every comparator above is locked to one family — Gaussian beliefs with Gaussian or Gumbel noise. The lattice’s base density is an array, so the noise model becomes an empirical question. We swept it.

base density	log loss	ECE	rank-PIT
$N(0, 1) + 25\%$ disaster slab	2.203	.015	.158
era-adaptive slab	2.200	.016	.185
Gaussian	2.225	.015	.176
skew-normal ($a=2$)	2.222	.010	.165
Student- t ($\nu=5$)	2.271	.018	.171
tight core (0.5) + 25% slab	2.254	.013	.136
near-Dirac core (0.2–0.3) + slab	2.42–2.52	—	—

The *disaster slab* — ordinary Gaussian pace with probability $1 - p$, a uniform lump of far-slow mass with probability p — matches the physics of retirement: a blown engine is not a slow day, it is a separated event that no level of skill outruns. It improves every metric.

Three hypotheses failed, and the pattern of failure is the finding. Symmetric heavy tails (t -family) hurt monotonically as tails fatten: symmetry forces miracle mass onto the *fast* side, and nobody outruns their ability by five sigma. Skew-normal noise does approximately nothing here, as it did on every dataset we tried; finish orders are blind to smooth asymmetry (pairwise comparisons cancel skew exactly, and ranks are invariant to monotone transformations of the clock). And a near-Dirac pace core — the hypothesis that finishing order among finishers is nearly deterministic — is refuted at 0.2 log loss: strategy, traffic and weather are order-one noise. A moderately tight core does, however, produce the best distributional calibration we measured (rank-PIT .136), so winner prediction and field-shape prediction pull the kernel in different directions: a genuine Pareto frontier, of which the table shows three corners.

5 The slab is physics

Fixing p at 25% is itself a Gaussian-era compromise. The era-adaptive variant estimates p prequentially from a trailing window of observed retirement frequency (no leakage: past races only) and rebuilds the base density on the fly. It sets the overall record (log loss 2.2001, best Kendall’s tau), and its fitted trajectory is the point: the mass falls across seven decades and ends near the modern retirement rate. The parameter is not a tuning constant; it is a measurement of the sport, made by the rating system in passing.

6 Against the betting market

No public archive of Formula 1 winner odds exists, so we built one: 523 Wayback Machine captures of odds-comparison pages yield full-field, multi-bookmaker consensus probabilities for 144 grands prix, 2008–2025 (overround removed; medians across roughly twenty bookmakers). On the 135 races past our warm-up, scored on identical matched-driver subsets:

	log loss
market consensus	1.547
tempered pool ($a=1.10$, $b=0.00$)	1.538
era-adaptive slab rater	2.117
uniform	3.018

The market sits 0.570 ± 0.065 of log loss ahead of the best fundamental rater we can build — a larger moat than we measured for the Hong Kong pari-mutuel — and the prequentially fitted pool assigns our ratings a weight of exactly zero. The explanation is not embarrassing: many archived snapshots postdate qualifying, so the market’s prices embed grid position, testing pace and weather — the present, where ratings compress only the past. Two caveats cut in opposite directions: snapshot timing favors the market (post-qualifying information), while snapshot staleness adds noise against it. The one exploitable wrinkle is familiar from every betting market: $a = 1.10$ says the odds are mildly underconfident in favorites, worth 0.008 — real, tiny, and far inside bookmaker margins. Ratings do not beat Formula 1 win markets. The open route is different: derivative markets (head-to-heads, top-six) are priced off these same win odds by Harville-style transforms that we have shown discard measurable structure; testing that requires derivative odds these archives lack.

7 What we did not do

Driver-versus-car identification is unaddressed: constructor effects are folded into driver ratings, and separating them (each driver a latent vector, the car a direction) is the natural next study. Retirements enter tied at the back, discarding the official ordering of retirees by laps completed; the alternative is an untested A/B. The derivative-market test above awaits data. And the ratings themselves are one-dimensional: no wet-weather axis, no circuit-type axis, though the machinery for both exists.

Reproducibility

Everything is one command per table from the `winning` package (github.com/microprediction/winning): `python -m winning.benchmarks.run_benchmark --dataset f1` for the benchmark, and `research/f1_dirac_`, `research/t_sweep.py`, `research/f1_era_slab.py` for the density studies. Data is fetched at run time from `f1db` (CC-BY-4.0). The underlying lattice machinery is the `thurstone` package; the ability transform is described in Cotton, *SIAM J. Financial Mathematics* 12(1), 2021.